

BATUHAN DORUK OZTURK

Full-Stack Developer | AI Pipelines & Web3

Istanbul, Turkey | +90 535 483 30 80 | batuhandoruk91@gmail.com

[GitHub](#) | [LinkedIn](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Full-stack TypeScript developer with 5+ years of professional experience and a strong specialization in AI-assisted systems and Web3. Built and shipped 10+ production applications across DeFi, fund administration, HR SaaS, loyalty platforms, e-commerce, and education. Core expertise in Node.js, Next.js, and React, with hands-on experience in distributed processing systems and smart contract development. Comfortable owning a product end to end with minimal oversight, from architecture through to final user experience. Applies more than a decade of engineering experience to identify product, user and technical risks early.

TECHNICAL SKILLS

Core Stack: Next.js | Node | React | TypeScript | PostgreSQL | Viem | Wagmi | Solidity

Languages: TypeScript, JavaScript, Solidity, HTML, CSS/Sass, SQL

Backend: Node.js, Express, Koa, Next.js API Routes, REST APIs, RPCs, GraphQL/Apollo, WebSocket, Redis job queues, signed webhooks, background workers

Frontend: React, Next.js, TanStack Query, Tailwind CSS, Material-UI, Headless UI, React Native, Expo

AI & LLM Engineering: OpenAI, DeepSeek, Gemini, Kimi; multi-provider fallback chains, prompt design, structured extraction, model benchmarking; Claude Code, Codex, Cursor, GitHub Copilot

Blockchain & Web3: Solidity, Foundry, Hardhat, OpenZeppelin, Viem, Wagmi, Ethers, Privy, Alchemy, ERC-4337, The Graph

Databases & ORMs: PostgreSQL, MySQL, MongoDB, Redis; Prisma, Sequelize, Mongoose

Infrastructure & Testing: Docker, GitHub Actions, NGINX, Vercel, Railway, Cloudflare R2, AWS S3, Supabase, Neon; Vitest, Jest, React Testing Library, Playwright, Axiom, PostHog, Sentry

Integrations: Stripe, Google APIs (Calendar, Docs, Maps, Wallet), Apple Wallet, Telegram Bot API, Resend, JWT, Passport.js, NextAuth, SAML/SSO (Azure AD, OKTA), OAuth, Zod, i18next

PROFESSIONAL EXPERIENCE

Full-Stack Engineer (Independent)

May 2026 – Present

Keylangan | Remote

Production platform that turns unstructured, free-form, human written descriptions into clean, deduplicated, searchable listings. Own the system end to end: the Next.js/PostgreSQL application, a standalone stateless extraction service with isolated network egress and signed callbacks, and the self-hosted deployment.

- Designed a **multi-stage processing pipeline** as modular service handlers, bounded by strict job contracts, handling heavy concurrent workloads while allowing each stage to be **changed or scaled independently**.
- Built **idempotent retries and automatic recovery** mechanisms for failure scenarios across external services and internal pipeline stages, preventing lost work and duplicate writes.
- Compared four LLM providers** on real data using accuracy, cost, latency, and output truncation metrics, then used the results to select a chain that cut **extraction cost 7x and latency 9x** while raising field accuracy.
- Diagnosed and fixed a **database egress problem** that scaled with table size, by pushing perceptual-hash distance filtering out of the application and into PostgreSQL at no loss of recall.
- Automated duplicate review using **image hashing**, banded locality-sensitive indexing, and a confidence-gated AI verdict layer handling several hundred decisions a day, **resolving roughly 96% without human review**.
- Ship the system as a **versioned Docker image built in CI**, deployable or reversible with a single command, covered by unit and integration tests against a disposable containerized PostgreSQL.
- Implemented PostHog **product analytics** and Axiom logging, then built custom event logs and operational dashboards to track pipeline throughput, queue health, and missing callbacks.

Full-Stack Web3 Developer

Jan 2024 – May 2026

Fume (via 5w155) | Remote

On-chain fund administration platform enabling crypto fund managers to create, operate, and report on tokenized investment vehicles across EVM chains (Ethereum, Arbitrum, Base, BSC).

- Designed and built **end-to-end transactional flows** that validate input, sign and dispatch write-contract calls, parse transaction receipts, and reconcile state back into PostgreSQL across signature, approval, deposit, and withdrawal paths.
- Architected a **multi-network, multi-wallet dApp** routing each operation to the correct chain and wallet, handling EOA and smart-account paths with **paymaster-sponsored gas via ERC-4337**.
- Contributed to **design decisions on the Solidity protocol** and its gas optimization passes, and implemented a **function-level onchain authorization system** controlled by wallet address, with matching admin settings UI.
- Implemented **subgraph indexing** across multiple linked contracts and block ranges, cutting event query latency by up to 5x and unlocking near-real-time UI updates.
- Drove **production reliability and 3x page load improvements** through targeted caching and Next.js SSR/ISR, guided by Sentry, Alchemy, and Vercel log analysis.

Full-Stack Developer

Jul 2022 – Dec 2023

5W155 | Remote

Delivered full-stack features across HR SaaS, multi-tenant loyalty, real estate, and mobile product lines.

- Architected a **configurable promotion and reward engine** for a multi-tenant loyalty platform, evaluating multiple criteria per cardholder (age, purchase frequency, birthday, card validity) on a Koa 2 / MySQL API.
- Integrated **Apple Wallet (PassKit) and Google Wallet** for digital loyalty cards, Stripe for merchant billing, and **QR-code payment generation**; built the React admin dashboard and the Expo mobile app.
- Built a **recruitment pipeline** with drag-and-drop status tracking, **Google Calendar integration** with auto-generated Meet links, and candidate CV generation via the **Google Docs batch-update API**.
- **Migrated a real estate quotation platform from Angular to React**, refactoring authentication and permissions, fixing price-formula bugs across property types, and extending the PDFTron generation pipeline.

Junior Full-Stack Developer

Jun 2021 – Jul 2022

LaSoft Agency | Lviv, Ukraine

Built features across 6 client projects in education, marketing, e-commerce, and health: **SSO/SAML authentication** (Azure AD, OKTA), WebSocket live chat, Stripe payments, Mixpanel dashboards, and SEO with Google structured data.

PERSONAL PROJECTS

Decentralized Lottery Protocol | Solidity, Foundry, OpenZeppelin, wagmi, viem | Solidity protocol (Governor + Lottery contracts) with fixed-fee entries, commission-based treasury sweeps, and winner selection via Chainlink VRF. Built the dApp frontend with wallet-connect flows and deployed on Arbitrum using Foundry.

EDUCATION

B.S. Civil Engineering

Graduated Aug 2015

Middle East Technical University (METU), Ankara, Turkey

5 years of post-graduation civil engineering and project administration experience (2016-2020), managing multi-site teams across Ukraine, Moldova, and Saudi Arabia.

ADDITIONAL

- **Languages:** English (fluent), Turkish (native)
- **Interests:** Swimming, Surfing, Diving, Chess, Pool, Board Games, Skiing, and numerous outdoor activities
- **References:** Pablo Pfister (CEO, 5w155), Marc Bickel (CTO, Fume) – contact details available on request